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REVIEWS

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PSGL-1: A Universal Selectin Ligand or a Signaling Molecule? (Review of the Literature)

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Abstract—Interactions of intercellular adhesion molecules of the selectin family with glycoconjugates of cell membranes mediate the initial stage of the adhesion cascade, which recruits leukocytes circulating in the bloodstream to sites of infection or damage. The formation of heterotypic cell aggregates between individual cells of hematopoietic and non-hematopoietic origin may be involved in processes leading to inflammation, thrombosis, and metastasis. A key protein, the dimeric glycoprotein PSGL-1, a P-selectin glycoprotein ligand, plays an important role in the binding of selectins, serving as a ligand for all three selectins. PSGL-1 combines signals activating various biochemical pathways during binding and rolling of leukocytes. The integration of these signals leads to activation of leukocytes, integrin-mediated arrest, restructuring of the cytoskeleton of interacting cells, polarization, and subsequent diapedesis of leukocytes into surrounding tissues. The multilevel effect of PSGL-1 on cellular traffic in the physiological and inflammatory states is largely determined by posttranslational modifications, among which an important place is given to specific O- and N-glycosylation and sulfation. In this review, we discuss modifications of PSGL-1 associated with the initiation of biochemical pathways, as well as its interactions, which make it possible to classify this molecule as signaling, paying special attention to the mechanisms leading to pathology, including cardiovascular.

Keywords: glycoprotein ligand of selectins, PSGL-1, adhesive function of the endothelium, selectins

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INTRODUCTION

Simple and complex carbohydrates, called glycans, constitute an important supramembrane complex of the apical surface of endothelial cells—the endothelial glycocalyx (EG), which mediates physiological and pathological processes [1]. Glycans have a three-dimensional structure, which determines their adaptation to perform physiological functions [2]. Normally, the components of intact EG provide protection against cell adhesion molecules, preventing pathological intercellular interactions; however, under normal conditions, blood cells interact both with each other and with endothelial cells. Glycans are capable of storing information in a volume greater than that contained in nucleic acids and proteins. This information is decoded by glycoproteins called lectins, in particular, molecules of the selectin family [3]. In the last 20 years, carbohydrate–protein specificity has been considered as an addition to the genetic code: glycoproteins, glycolipids, polysaccharides, and proteoglycans have the highest potential for encoding biological information. In carbohydrate complexes, information can be encoded not only by the sequence of monomers, which is typical of protein molecules, but also by the anomeric configuration and position of bonds, as a result of which carbohydrate chains have unique coding capabilities.

Selectins are cell adhesion receptors that contain Ca^{2+}-dependent glycan recognition and binding domains. There are three known representatives of this family, P, E, and L-selectins, which mediate the initial stages of leukocyte binding to the damaged endothelium with the prospect of subsequent rolling and transvasation into foci of inflammation and their emigration into lymphoid tissues. Cell attachment and rolling on the endothelial surface occurs in the vascular bed under the influence of shear stress, which requires selectins to have unique biophysical properties. Having high binding kinetics, selectins facilitate the formation of temporary connections and a dynamic contact zone that moves along the vascular endothelium during cell rotation [4].

Currently, the participation of selectins in the immune response, hemostasis, and tissue healing process, as well as in inflammatory reactions and carcinogenesis has been discovered, in connection with which their contact with ligands may also have therapeutic significance [5].

It is known that a large number of different molecules bind to selectins. However, not all of them are

involved in biological processes. In this regard, a number of characteristics have been put forward that a true selectin ligand must meet. These include the following:

(1) The ligand molecule must be present, as they say, in the right place at the right time; for example, the true ligand of P-selectin must be on the surface of a mature circulating leukocyte, when P-selectin itself is actively expressed on the surface of endothelial cells or platelets in the vascular bed.

(2) Inactivation of the putative ligand on an inactivated cell excludes biochemically appropriate interactions.

(3) Each selectin binds to the ligand with high, but not equal, affinity and with a certain stoichiometry [6].

As it turned out later, the PSGL-1 protein meets all the requirements listed (*P-selectin glycoprotein ligand 1; SELPLG; CD162; CLA; PSGL-1; PSGL1*).

THE STRUCTURE OF PSGL-1

The search for a ligand for the first of three known selectins, P-selectin, led to the discovery of PSGL-1 in neutrophils in 1993. PSGL-1 is a transmembrane bitopic glycoprotein type I, expressed in lipid rafts at the apex of leukocyte microvilli [7]. PSGL-1 was initially identified in the human neutrophil-like cells HL-60 (human leukemia cell). It is now known that PSGL-1 is expressed by all leukocytes: neutrophils, monocytes, and most lymphocytes (including B cells, natural killer cells, naive cells, and memory T cells), as well as platelets and some endothelial cells. In addition, there is evidence of the presence of PSGL-1 in chronically inflamed human prostate tissue, on activated human umbilical vein endothelial cells (HUVEC), and on endothelial cells of the small intestine and mesenteric lymph nodes of mice with chronic ileitis [8, 9]. PSGL-1 is a homodimer, the monomers of which are linked by a disulfide bond. The molecular mass of the monomer is 120 kDa. The PSGL-1 precursor has a molecular mass of 43.2 kDa and consists of 412 amino acid residues: a signal peptide (1–17) and a propeptide (18–412). It includes three domains: N-terminal extracellular (303 residues, 18–320), transmembrane (21 residues, 321–341), and short C-terminal cytoplasmic (71 amino acid residues, 342–412). PSGL-1 is activated by partial proteolysis: the mature protein consists of 371 amino acid residues. After final maturation and glycosylation, the PSGL-1 dimer has a molecular mass of 240 kDa.

The extracellular domain, the N-terminal region of the protein, 50–60 nm in length [10], contains branching sites at which O- and N-glycosylation occurs, and terminal sites that undergo final modifications leading to the ability to bind to selectins. PSGL-1 is enriched with amino acid residues such as Thr, Pro, and Ser. The N-terminal region also contains three Tyr residues at positions 46, 48, and 51, which undergo sulfation, and 14–16 decameric repeats carrying O-linked glycans [11]. The amino acid sequences of the transmembrane and cytoplasmic domains are thought to be highly conserved, implying important functions.

The extracellular domain contains numerous glycans linked to the protein by O-glycosidic bonds at the Ser and Thr residues. Near the transmembrane domain, there is a Cys residue that forms a disulfide bond, which is thought to stabilize the dimeric form of the protein.

The short cytoplasmic tail of the PSGL-1 molecule contains a juxtamembrane cytoplasmic sequence of 18 amino acids (RLSRKGHMYPVRNYSPTE in humans) [12]. Conserved residues in the cytoplasmic region of PSGL-1 may bind ezrin and moesin, members of the ezrin/radixin/moesin (ERM) family of adaptor proteins responsible for membrane-cytoskeletal connections [13, 14].

POST-TRANSLATIONAL MODIFICATIONS OF PSGL-1

The functioning of the protein does not have a clear dependence on the expression level. For the normal functioning of the PSGL-1 ligand, important post-translational modifications are necessary. These modifications include the following transformations: glycosylation [15], tyrosine sulfation [16, 17], and the formation of branched carbohydrate side chains with the participation of the core protein [18]. The key O-glycan in PSGL-1 containing the smallest recognition epitope is the tetrasaccharide sialyl Lewis X (sLe^x)), and its analogue sialyl Lewis A (sLe^A), located near the N-terminus in human PSGL-1 at Thr-57. All selectins recognize sLe^X; however, only one sLe^X may be sufficient for interactions with selectin E. The other two selectins require a ligand modified by additional sulfation, such as GlcNAc-6-sulfation of L-selectin; in selectin P, in addition to sLe^X, at least one additional O-sulfated tyrosine residue of the polypeptide chain is required. Such modifications of ligands increase the affinity of their binding to selectins. The carbohydrate structures that are determinants of selectin binding are formed as a result of post-translational modifications of proteins or lipid scaffolds using enzymes required for glycosylation—glycosyltransferases, enzymes that catalyze the transfer of a monosaccharide residue from a donor to the hydroxyl group of an acceptor, thereby forming a new glycosidic bond. There is reason to believe that for each glycosidic bond there is its own glycosyltransferase [19]. The donors in this reaction are nucleotide sugars, such as UDP-GlcNAc, UDP-GalNAc, UDP-Glc, UDP-Gal, GDP-Fuc, GDP-Man, and CMP-NeuAc. The side groups of synthesized glycans can be chemically "modified"; for example, sulfotransferase modifies the OH6 hydroxyl

Table 1. Enzymes involved in post-translational modifications of PSGL-1

Enzyme	Gene name	Modification
Fucosyltansferase IV or VII	*FUT4* or *FUT7*	α-1,3-fucosylation
Core 2-β-1,6-glucosaminyltransferase-I	*GCNT1*	formation of branched carbohydrate side chains
β-1,4-galactosyltransferase-I	*B4WRONG1*	β-1,4-galactosylation
Sialyl 3-transferase IV	*ST3GAL4*	α-1,3-sialylation
Tyrosylprotein-O-sulfotransferase 1 or 2	*TPST1* or *TPST2*	tyrosine-O-sulfation

of GlcNAc by adding a sulfate group and forms 6-sulfo-sLeX, the ligand of L-selectin.

Enzymes that actively participate in post-translational modifications of PSGL-1 are presented in Table 1.

Post-translational processing of PSGL-1 involves sulfation of N-terminal tyrosine residues, which can be performed by at least one of two tyrosine sulfotransferases. Myeloid cells are known to express enzymes required for PSGL-1 glycosylation constitutively. In contrast, $CD4^+$ T cells synthesize significant amounts of PSGL-1 but lack fucosyltansferase VII and 2-β-1,6-glucosaminyltransferase-I C2GlcNAcT-I, which are necessary for the appropriate modification that can provide binding to P-selectin [20]. In addition, sialyl 3-transferase IV is expressed at low levels. Naive T cells proliferate and differentiate only upon presentation of the cognate antigen. Expression of the above enzymes is induced immediately after differentiation of T cells into T helper type I (ThI) cells and their activation, which leads to modification of PSGL-1 for successful binding to selectins; i.e., the ability to bind to selectins is acquired only during proliferation and differentiation of effector T cells [21]. In naive T cells, the incompletely glycosylated protein is able to bind the structurally homologous chemokines CCL21 and CCL19, which stimulate leukocyte migration to secondary lymphoid organs through an enhanced chemotactic T cell response.

Thus, among the post-translational modifications of PSGL-1, the most important for binding are α-1,3-fucosylation, α-2,3-sialylation, β-1,4-galactosylation, and 2β-1,6-N-glycosylation by acetylglucosaminyltransferase.

INTERACTION OF PSGL-1 WITH THE CELL CYTOSKELETON

PSGL-1 is capable of performing an adhesive function only if it is associated with the cell cytoskeleton [22]. Cells expressing PSGL-1 lacking the intracellular domain, as well as cells with actin cytoskeleton damaged by pharmacological agents, were characterized by a significant decrease in rolling efficiency. As noted above, the cytoplasmic domain of PSGL-1 is linked to the actin cytoskeleton via adapter molecules, the proteins ezrin, radixin, and moesin (ERM). These ERM proteins regulate the redistribution of PSGL-1 across the cell plasma membrane from the tips of leukocyte microvilli to the posterior projections of polarized cells, the uropods. ERM phosphorylation results in enhanced translocation of PSGL-1 to T cell uropods. After binding to ERM proteins, PSGL-1 is associated with the cell's actin cytoskeleton and with signaling molecules such as MAPKs (mitogen-activated protein kinases), the functions of which in cells are diverse and include control of gene transcription, metabolism, proliferation, motility, and cell apoptosis.

In addition, PSGL-1 is associated with the FERM domain of radixin (found in ERM proteins), which consists of three subdomains: A, B, C. Interaction with PSGL-1 is via the C-domain; the β-chain of PSGL-1 forms an antiparallel β-β-structure with the β5C chain of subdomain C, and 12 amino acid residues are involved in the formation of hydrogen bonds. The PSGL-1 binding site in the FERM domain of radixin overlaps with the binding site of the adhesion molecule ICAM-2 (Intercellular adhesion molecule 2, CD102). The ITAM-like motif (immunoreceptor tyrosine-based activation motif, ITAM, amino acid residues 191–208) acts as a hinge region linking subdomains B and C, and two phosphorylation sites Y191 and Y205 are hidden by subdomains B and C, respectively. This incorporation of the Y191 and Y205 sites suggests that a steric hindrance effect exists both through the action of the phosphorylation sites on the ITAM-like motif and through the recruitment of the nonreceptor spleen tyrosine kinase Syk, preventing excessive leukocyte activation via the PSGL-1 signaling pathway under hemodynamic conditions [23].

The nine PSGL-1 residues that comprise the nonpolar region directly contact subdomain C. As in ICAM-2, the conserved nonpolar residues Tyr10 and Val12 of PSGL-1 are located in a large hydrophobic pocket formed by Ile245, Ile248, Leu281, His288, and Met285. In addition, Tyr10, like His288, forms a hydrogen bond in both proteins.

There is another highly conserved cell membrane–associated protein that localizes to lipid rafts, flotillin-2 (Flot-2), which may also play a role in recruiting PSGL-1 to uropods in T cells, thereby promoting their activation [24]. In addition, Flot-2 interacts with PAR-1, a mediator of key signaling pathways involved

in cell growth and metastasis, and may promote tumor progression.

In addition to ERM proteins, the PSGL-1 molecule is associated by its cytoplasmic domain with the spleen cytoplasmic nonreceptor tyrosine kinase Syk and the metalloproteinase ADAM8, with ERMs acting as adapters in the latter case. A. Urzainqui et al. showed that the interaction of PSGL-1 with P-selectin leads to phosphorylation of Syk and direct effects on the amino acid sequence of the N-terminal region of the immunoreceptor, which is the site of ITAM activation in moesin or ezrin in human T-lymphoblasts. These proteins are then phosphorylated at tyrosine residues, which leads to the formation of the PSGL-1-Syk-moesin complex and, ultimately, to the activation of the proto-oncogene c-*Fos* [25]. The association of PSGL-1 with an enzyme belonging to the ADAM8 family of metalloproteinases causes cleavage of the extracellular domain of PSGL-1 and blocks neutrophil rolling. At the same time, the association of the PSGL-1 decameric repeat domain with ADAM28 enhances binding to P-selectin, which provides regulation of the recruitment of immune cells to sites of inflammation through the binding of P-selectin to its ligand PSGL-1 [26].

M. Matsumoto and T. Hirata showed using intravital microscopy that moesin can be partially involved in PSGL-1-mediated activation of β2-integrin and slowing down of the leukocyte rolling velocity after binding of P-selectin to PSGL [27]. Thus, in mice with a knockout of the moesin gene, a slight increase in the rotation rate of neutrophils in the venules of the cremaster muscle is observed when exposed to tumor necrosis factor α (TNF-α) [27]. The neutrophil binding observed with decreased moesin synthesis may be due to a compensatory interaction of PSGL-1 with ezrin. PSGL-1, by interacting with ERM proteins through phosphorylation of extracellular signal-regulated kinase (ERK), is involved in the activation of the MAPK kinase pathway, which is not associated with Syk phosphorylation. It has been noted that activation of two pathways, ERK and Syk, induced by PSGL-1 depends on the integrity of lipid rafts [28]. The role of lipid rafts in PSGL-1 redistribution is also being studied. Thus, T. Xu et al. showed that the integrity of the lipid raft is necessary for PSGL-1-mediated "rolling" of neutrophils on P-selectin. Ligation of PSGL-1 in HL-60 cells causes rapid phosphorylation of the β-adducin protein and mediates ligation-dependent distribution of PSGL-1 by dissociation from the actin cytoskeleton [29].

PSGL-1 BINDING TO SELECTINS P, E, AND L

As already mentioned, PSGL-1 is expressed on the surface of lymphoid and myeloid cells, including platelets and activated T cells and through interactions with selectins is involved in the binding of leukocytes to endothelial cells, facilitating subsequent extravasation into tissues. Currently, PSGL-1 is recognized as the best studied physiological ligand, which satisfies all previously proposed criteria for a true ligand. It is able to bind with high affinity through the N-terminal region of the polypeptide chain to all three selectins, but its affinity depends on the type of selectin: K_D P-selectin is 320 nM, E-selectin is 15 μM, and L-selectin is 5 μM [30].

When describing the interactions of PSGL-1 with selectins, it is necessary to return to the structure of the extracellular domain, which is endowed with conformational flexibility that facilitates binding to counter-receptors. This domain contains Ser and Thr residues and Pro-rich decameric repeats that vary among species and protrude above the cell surface. In general, the extracellular domain, unlike the transmembrane domain, may have some variability. However, despite this, the O-glycosylation site Thr58, located near the N-terminus, is conserved. It is this domain that carries sLe^X and its analogue sLe^A, which communicates with selectins. Tyrosine sulfation (Tyr46, Tyr48, and Tyr51) provides high affinity binding of human PSGL-1 to selectins. In addition, sulfated tyrosine promotes the binding of the chemokines CCL19 and CCL21 to the N-terminus of PSGL-1, whereas branching at the O-glycan attachment points and sialylation reduce binding [31]. The key event in selectin binding is electrostatic interactions between Ca^{2+} ions and hydroxyl groups at the C3 and C4 atoms of the fucose residue in the tetrasaccharide sLe^X. Formation of hydrogen bonds between the galactose residue sLe^X and Tyr4 and Glu92, between sialic acid and Tyr48, provides stabilization of the selectin-ligand complex [32]. Following ligation with selectins, PSGL-1 localizes to membrane microdomains called microvilli at the trailing edge of migrating leukocytes, placing it in close proximity to other adhesion and signaling molecules such as CD43 and CD44 and supporting leukocyte attachment to the endothelium.

A. Leppanen et al. investigated the role of tyrosine sulfation by synthesizing glucosulfopeptides that model the PSGL-1 binding region [33]. It has been established that binding to P-selectin requires sulfation of at least one tyrosine, and sulfation of Tyr48 (TO_D ~ 6 μM) promotes higher affinity than Tyr46 sulfation (TO_D ~ 10 μM) or Tyr51 (TO_D ~ 10 μM). The model with all sulfated tyrosine residues (Tyr46, Tyr48, and Tyr51) showed a higher binding affinity (TO_D = 0.65 μM), while the nonsulfated model had a significantly lower affinity (TO_D ~ 25 μM). This study also showed that the contribution of fucose to binding is more significant than that of sialic acid, and all interactions are charge dependent. It has also been shown that sulfation of the GlcNAC residue by sLe^X leads to the formation of 6-sulfo-sLe^X, which increases the binding affinity for L-selectin. Moreover, recognition and binding to E-selectin do not depend on sulfation [34].

V. Sladek et al. [35] showed that mutations in the E-, L-, and P-selectins, for example, at positions 46, 85, 97, and 107, affect ligand functioning differently. Furthermore, it is suggested that sulfation of the tyrosine residue at position 51 leads to interaction with Arg85 in P-selectin, which is a significant factor in the increased affinity of P-selectin for PSGL-1 compared to the E- and L-selectins [35].

It is now known that single nucleotide or complex polymorphisms in the selectin and PSGL-1 genes may be the cause of differences in the structure of these proteins, which may lead to functional effects and the development of vascular and metabolic diseases. Thus, polymorphism of the number of tandem repeats is relatively common (*variable number of tandem repeats, VNTR*), which affects the length of the extracellular domain of the PSGL-1 molecule, including the distance from the P-selectin binding site to the cell surface. In humans, three allelic variants are known: A, B, and C, which contain 16, 15, and 14 decameric repeats, respectively. Allele B does not have repeat 2; C contains this repeat, but its structure lacks repeats 9 and 10. The highest frequencies were for the A allele, and the lowest, for C. Homozygous carriers of the short B and C alleles had a lower risk of premature myocardial infarction due to lower adhesive capacity [36].

The decameric repeats contain repeated regions of ten amino acids containing approximately 30% O-glycated threonine and 10% proline residues, which lengthen and strengthen the protein scaffold and separate the N-terminal selectin binding sites from the cell membrane. Glycosylated decamers account for nearly two-thirds of the molecular mass of PSGL-1 and contain at least 50 potential O-glycosylation sites. Deletions in genes encoding decameric repeats disrupt P- and, especially, L-selectin-dependent interactions with PSGL-1 [37]. In studying recombinant decamers, S. Tauxe et al. showed for the first time that decamers directly interact with E-selectin, unlike P-selectin, and provide up to 80% of leukocyte rolling on E-selectin. They found that the main function of decamers is to extend the N-terminus of PSGL-1 and "remove" it from the cell surface to expose it to flowing leukocytes, activated endothelial cells and platelets, and adherent platelets or leukocytes, and to limit the flexibility of the peptide backbone. They also enhance the ability of the PSGL-1 N-terminus to support L- and P-selectin-dependent binding, play a major role in stabilizing L-selectin-dependent rolling interactions, directly interact with E-selectin, and act as an adhesion receptor [38].

PSGL-1-TRIGGERED SIGNAL CASCADES

In addition to its adhesive function, PSGL-1 is characterized by its participation in the initiation of a number of signaling cascades. Nonreceptor tyrosine kinases of the Src family phosphorylate a variety of cytosolic, nuclear, and membrane proteins and play a central role in PSGL-1-mediated signaling. Disruption of their functioning can contribute to the progression of cellular transformation and oncogenic activity. A special role is played by the already mentioned nonreceptor tyrosine kinase Syk, which, when activated by phosphorylation, binds to the ITAM-like motif in moesin and thus indirectly binds to PSGL-1 and induces transcription of the early response gene c-*Fos*. c-Fos, in turn, is part of the transcription factor activator protein-1, which regulates the expression of other cell adhesion molecules—ICAM-1 and VCAM, which are necessary at the next stages of the adhesion cascade. Thus, by controlling the expression of ICAM-1 and VCAM-1, c-Fos also controls transendothelial migration of leukocytes [39].

PSGL-1 has also been shown to be required for G protein-coupled receptor-mediated independent activation of integrins. Continuous binding of PSGL-1 to E-selectin induces signaling events that increase the affinity of LFA-1 (integrin αLβ2; *Lymphocyte function-associated antigen 1*) to ICAM-1 and mediates slower binding of ICAM-1 to E-selectin.

Recently, a new regulatory mechanism has been identified in which cytosolic myeloid proteins (MRP) 8 and 14, secreted by myeloid cells, as well as PSGL-1, are used as proinflammatory mediators [40]. E-selectin-PSGL-1 interaction has been shown to trigger the release of Mrp8/14 and, in turn, activate the small GTPase Rap1 by Toll-like receptor 4 (TLR4) [40]. This results in rapid activation of β2 integrins. As a consequence, this pathway enhances the adhesion and transendothelial migration of leukocytes. Binding of P-selectin to its ligand also enhances integrin-dependent activation and adhesion to the immobilized ICAM-1 molecule.

It is assumed that, acting through PSGL-1, P-selectin induces the activation of Src kinase, which affects the processes of cellular differentiation, proliferation, and adhesion, the disruption of regulation of which can lead to the progression of cellular transformation and oncogenic activity; hyperactivation of PI3 is also associated with the development of oncopathology. This, in turn, leads to the activation of macrophage antigen-1 (Mac-1, αMβ2, CD11b/CD18, CR3) and LFA-1 on leukocytes [41].

PSGL-1 INVOLVEMENT IN PATHOLOGY

It is known that selectins and PSGL-1 are involved in physiological processes such as hemostasis, immune response, and wound healing. They are also involved in the processes of metastasis and, consequently, in the pathogenesis of oncological diseases. Over the past three decades, knowledge about the functions of selectins and PSGL-1 in the development and progression of a number of pathologies has expanded significantly.

Thus, the study of cellular interactions in inflammatory processes continues. For example, it has been shown that PSGL-1-P-selectin interaction through effects on the Syk/Ca^{2+}/PAD4 signaling pathway induces extracellular neutrophil trap formation in acute pancreatitis. Interfering with selectin-PSGL-1 interactions in a mouse model of acute pancreatitis reduces the severity of histopathological manifestations, which may be a promising strategy for the treatment of this pathology [42].

The involvement of PSGL-1 and selectins in cardiovascular pathology is actively being studied. Thus, in an experiment on mice, it was shown that suppression of GALNT4, one of the enzymes involved in post-translational O-glycosylation of PSGL-1, reduces atherosclerotic plaque formation, adhesion, and transmigration of monocytes by the Akt/mTOR and NF-κB pathways. Thus, GALNT4 may also be considered as a potential therapeutic target in atherosclerosis [43].

Furthermore, co-signaling of PSGL-1 and CXCR2 increases the incidence of thrombus and size in mice, promoting the development of deep vein thrombosis, in part by stimulating the release of neutrophil extracellular traps. Unlike neutrophils, blocking PSGL-1 or CXCR2 signaling in monocytes did not affect their recruitment to thrombi or tissue factor expression [44].

Increased expression of PSGL-1 was found in patients with aortic aneurysm. Moreover, PSGL-1 deficiency significantly reduced the incidence and severity of aortic aneurysm, and also reduced elastin fragmentation, collagen accumulation, and smooth muscle cell degeneration. This protective effect of PSGL-1 inhibition was mediated by a decrease in adhesion molecule levels and a subsequent reduction in leukocyte adhesion to the endothelium through the NF-κB pathway, ultimately leading to a decrease in inflammatory cell infiltration and expression of inflammatory factors [45].

It has been established that, when neutrophil activation is caused by gene knockout *KLF2* (*Kruppel-like factor, KLF2*) or the introduction of antiphospholipid antibodies (APS), clustering of PSGL-1 occurs through remodeling of cortical actin, thereby enhancing adhesion at the sites of thrombosis. The use of nanoparticles with PSGL-1 antibodies recapitulates the situation with PSGL-1 clustering and results in a reduction in thrombotic manifestations, demonstrating the importance and feasibility of targeting activated neutrophils to prevent immunothrombosis [46].

Attempts have also been made to study the involvement of PSGL-1 in the pathogenesis of vascular diseases with the aim of further searching for mechanisms regulating its interactions with selectins. Differences were found in the concentration of this PSGL-1 in the blood serum of healthy donors and patients with vascular diseases of the lower extremities such as varicose veins, acute venous thrombosis, and peripheral atherosclerosis [47–49].

R. González-Tajuelo et al. [50] studied the effect of PSGL-1 on pulmonary vascular wall remodeling and concluded that PSGL-1 deficiency in leukocytes is associated with a decrease in the number of regulatory T cells (Treg), NO production, and estrogen receptor (ERα) expression, followed by an increase in angiotensin II levels in the lungs of female mice, which contributes to the development of pulmonary arterial hypertension [50]. The absence of PSGL-1 promotes endothelial dysfunction characterized by a decreased vasodilation response due to impaired NO generation caused by decreased phosphorylation of endothelial NO synthase at the Ser^{1176} activation site.

The role of PSGL-1 in the immune response continues to be studied. Thus, the adverse effect of this ligand on the course of infection caused by HIV has been described. PSGL-1 has been shown to inhibit the process of reverse transcription of HIV RNA and inhibit the infectivity of HIV virions produced by cells expressing PSGL-1. It has been established that PSGL-1 expression negatively affects the functions of immune cells, especially T cells, which are the most important participants in protection against HIV infection. Indeed, PSGL-1 expression and signaling induce T cell exhaustion. In addition, PSGL-1 can also mediate the uptake of viral particles and their transfer into cells. Thus, PSGL-1 negatively affects immune cells during HIV infection and may potentially predict its progression [51].

The possible involvement of PSGL-1 in the pathogenesis of the new coronavirus infection COVID-19 was also studied. The results of this study revealed increased activation of the P-selectin-PSGL-1 pair with increased efficiency of primary leukocyte recruitment, which promotes tissue damage and immunothrombosis. It was concluded that endothelial cell activation and excessive leukocyte migration involving the P-selectin-PSGL-1 pair play a central role in the pathogenesis of COVID-19 [52].

CONCLUSIONS

Disruption of interactions between selectins and their main ligand PSGL-1 is observed in inflammatory diseases, arterial and venous thrombosis, atherosclerosis, and changes in the immune response. Due to their active involvement in such serious pathologies, the interaction of selectins with PSGL-1 becomes a therapeutic target in such conditions. PSGL-1 is considered in a broader sense: it is not just a ligand for adhesion molecules, but a signaling molecule capable of triggering a number of biochemical cascades. In this regard, it is relevant to search for the mechanisms underlying the regulation of the activity of interactions of cell adhesion molecules with each other and with their ligands, and PSGL-1 can serve as a marker mol-

ecule, targeting which in the future may have therapeutic application.

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CONFLICT OF INTEREST

The authors of this work declare that they have no conflicts of interest.

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